

Arduino-Controlled Namaste Greeting Robot



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The concept of controlling various servo motors through Arduino Uno board is introduced here with a fun project called 'namaste robot.' The robot turns its head by 180° and scans people in its range using an ultrasonic module. If it finds anyone nearby, it greets the person with 'namaste' with both hands pressing to-

gether, which is the traditional Indian way of wishing people. The robot can be used in offices, shopping centres, parks and party halls where it can greet and attract people.

Fig. 1 shows the author's prototype of this robot. Please note that the ultrasonic module is mounted on the chest in author's prototype, but during testing it was observed that if it is mounted on the eyes, it can scan a much larger area in its vicinity.

Circuit and working

Fig. 2 shows the circuit diagram of namaste robot. Servo motors M1

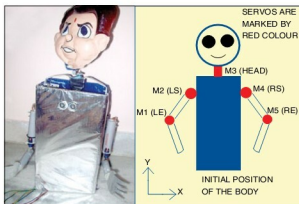


Fig. 1: Author's prototype of the namaste greeting robot

Test Points

Test point	Details
TP0	0V, GND
TP1	5V (with respect to TP0)
TP2	6V (with respect to TP0)
TP3	High pulse corresponding to the distance of obstacle

PARTS LIST

Semi-conductors:

- IC1 - APR9301-V2 voice recorder and playback IC
IC2 - 7806, +6V regulator
LED1, LED2 - 5mm LED
T1 - BC547 npn transistor

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 52-kilo-ohm
R2, R3 - 680-ohm
R4 - 15-kilo-ohm
R5, R7 - 10-kilo-ohm
R6 - 100-kilo-ohm
VR1 - 10-kilo-ohm preset

Capacitors:

- C1, C4 - 0.01µF ceramic disk
C2 - 22µF, 16V electrolytic
C3, C6, C7 - 0.1µF ceramic disk
C5 - 100µF, 25V electrolytic

Miscellaneous:

- BATT.1, BATT.2 - 9V battery
BOARD1 - Arduino Uno board
CON1, CON3 - 4-pin connector
CON2 - 5-pin connector
LS1 - 8-ohm, 0.5W speaker
M1-M5 - 5V servo motor
MIC1 - Electret microphone
S1 - Push-to-on switch
US-MODULE - HC-SR04 ultrasonic transceiver module

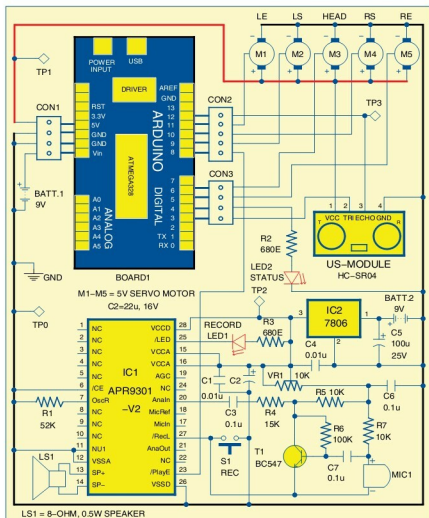


Fig. 2: Circuit diagram of the Arduino-controlled namaste greeting robot